

ABHINAV KAUSHIK

Senior AI Engineer

Bengaluru, India | +91 87005 71859 | abhinav.kaushik@gmail.com

SUMMARY

Senior AI/ML Engineer with 8+ years of experience delivering production AI and machine learning solutions across Healthcare, Telecom, AIOps, and client-facing technology environments. Holds an MSc in Machine Learning and AI from Liverpool John Moores University. Skilled in fine-tuning large language models and building end-to-end AI solutions for enterprise customers, with contributions to open-source LLM tooling and agentic AI frameworks. Experienced across the full lifecycle from rapid prototypes and proofs-of-concept to production deployment using GenAI, RAG, LangGraph, NLP, predictive analytics, anomaly detection, and time-series forecasting. Also designed UI/UX experiences, translating business and client requirements into practical AI solutions, and built a non-profit AI ed-tech platform for people with ADHD.

TECHNICAL SKILLS

AI / GenAI: GenAI, LLM, RAG, Agentic AI, LangChain, LangGraph, Claude, OpenAI, Embeddings, Tool Orchestration, Cross-Encoder, Transformers, NLP, Generative AI, LLM fine-tuning

Machine Learning: XGBoost, LightGBM, Random Forest, LSTM, SBERT, DBSCAN, Time Series Forecasting, Demand Forecasting, Anomaly Detection, Statistical Modeling, Regression, Deep Learning, PyTorch

Programming / Data: Python, SQL, PySpark, Pandas, NumPy, Flask, Hive, SparkQL, REST API, Pydantic, Unix
MLOps / Cloud: MLflow, Docker, Kubernetes, Jenkins, Bamboo, Bitbucket, DVC, Terraform, Airflow, GCP, Azure, NCI Servers

Search / Observability: Elasticsearch, Kibana, Logging, Log Analysis, Apache Solr, Lucene Indexing, Search and Information Retrieval

Data Platforms: Databricks, Apache Spark, Airflow

Databases: TimescaleDB, MySQL, Vector DB, pgvector, ArangoDB, Cosmos DB

BI / Testing: Power BI, QlikSense, SAP HANA, Selenium

Additional: Open-source contribution

PROFESSIONAL EXPERIENCE

Senior AI Engineer | Optum.AI

Bengaluru, India | Nov 2025 - Present

- Built and integrated a lab extraction agent with an existing Laboratory Information Management System (LIMS) using Claude Opus 4.7 and Amazon Textract, automating work associated with 800 Jira tickets and PRD tasks.
- Engineered a LangGraph-based analytics system for AARP medical plans, combining stateful multi-agent reasoning with real-time chart generation and reducing manual analysis effort by 98%.
- Enhanced dense retrieval performance by training and evaluating a cross-encoder with positive and negative examples for AARP survey data, achieving a 20% increase in retrieval accuracy.
- Led pre-sales technical workshops with enterprise clients, translating business requirements into AI solution prototypes and demos.
- Fine-tuned Claude Opus models on domain-specific medical data to improve extraction accuracy and reduce manual effort.

Senior Machine Learning Engineer | Viavi Solutions

Chennai, India | Sep 2024 - Nov 2025

- Built 30+ AI application prototypes and mockups for clients across AIOps and GeoPerformance, translating client requirements into functional POCs for solution validation and demonstrations.
- Facilitated development of an AI-powered Project Discovery Assistant that analyzes client transcripts, summaries, and uploaded documents using LangChain and LangGraph, achieving a 90% match rate of technical use cases with existing organizational solutions.
- Deployed a production-grade SLA breach prediction system using Random Forest and LightGBM with FastAPI and MLflow, integrating GPT-4-powered reasoning for real-time root-cause analysis and remediation guidance, resulting in a 67% reduction in downtime.
- Collaborated with product and research teams to integrate state-of-the-art LLM inference and fine-tuning pipelines into customer-facing applications.
- Authored and released open-source LangGraph extensions for multi-agent orchestration, gaining community adoption.

Data Scientist | Viavi Solutions

Bengaluru, India | Nov 2021 - Sep 2024

- Applied Sentence-BERT and DBSCAN to enhance semantic understanding and cluster alarm messages across graph-based AIOps infrastructure, achieving a 73.5% improvement in semantic alarm classification.
- Applied Isolation Forest and Holt-Winters for anomaly detection and demand forecasting across 500+ telecommunications network devices using 15+ KPIs, supporting earlier identification of abnormal time-series behavior and proactive network planning.

- Delivered technical POCs and consulting sessions on data services and products for 15+ clients across 3+ AIOps and ML platforms, contributing to an 11% increase in POC and mentoring revenue.

Data Analyst | United Health Group

Gurugram, India | Jun 2021 - Nov 2021

- Collaborated with stakeholders on ad-hoc analysis and customer-retention dashboards using Hive, SQL, and Power BI.
- Developed a Python- and Flask-based parser to automate data extraction, parsing, and transformation into structured data, supporting downstream analytics and reporting.

Data Engineer | Nippon Data Systems

Gurugram, India | Nov 2019 - May 2021

- Engineered SQL and PySpark-based data pipelines to automate BAU medical-audit reporting, transforming and aggregating large healthcare datasets and reducing manual reporting effort by 95%.
- Built real-time observability using Elasticsearch for centralized log ingestion and analysis, with Kibana dashboards for monitoring system activity and user events, improving traceability efficiency by 80% for the COVID QRT team.
- Developed a Flask-based enterprise search engine using Apache Solr and Lucene indexing, enabling centralized search and retrieval across organizational data for enterprise-wide use cases.

Associate Consultant | Saksoft

Gurugram, India | Dec 2017 - Nov 2019

- Conceptualized customer dashboards using QlikSense, SQL, and SAP HANA for financial and product insights, resulting in a 40% increase in user engagement.
- Developed an XGBoost-based propensity model to predict the probability of customers initiating a personal loan application within 7 days of contact, enabling targeted outreach and customer prioritization.
- Leveraged Selenium and Python to automate performance testing and evaluate application response times, resulting in a 90% reduction in testing effort.

PROJECTS

Think Tree - ML Study Platform | Live Platform | <https://think-tree-ai.vercel.app/>

- Built a non-profit AI-powered ML study platform for people with ADHD and context-switching challenges, designed to help users organize learning and maintain study context.
- Developed RESTful FastAPI endpoints and a web UI to deliver personalized learning paths powered by LLMs.
- Integrated vector-search with pgvector to enable semantic content retrieval for study materials.

Financial Fraud Detection | Mar 2025 | GitHub

- Developed an AI-driven fraud detection and reasoning workflow using the JPMorgan Synthetic Dataset, combining LightGBM-based modeling with LLM explanations to support interpretable financial-risk analysis.

Restaurant Feedback Analyzer - Aspect-Based Sentiment Analysis | GitHub

- Built an NLP pipeline to extract review aspects and determine sentiment at the aspect level, enabling granular insights into customer experience.

RFP Multi-Agent Proposal Automation Platform | GitHub

- Built a multi-agent RFP automation platform using an LLM-driven supervisor to orchestrate intent detection, retrieval, proposal generation, document matching, and market research with RAG over historical proposals.
- Implemented fine-tuning of an LLM supervisor on historical proposal data to improve generation relevance.
- Published core RAG and agentic workflow components to an open-source repository, receiving community contributions.

EDUCATION

Master of Science, Machine Learning and AI | Liverpool John Moores University

Jan 2020 - Jan 2022

Bachelor of Technology, Information Technology | Jamia Hamdard

Aug 2013 - Aug 2017

Relevant coursework: Deep Learning, Natural Language Processing, Model Deployment and Scaling

ADDITIONAL INFORMATION

- German: A1 (in progress)
- Network Intern, Airport Authority of India
- Hadoop Intern, Ducat Technologies
- Think Tree - built an AI-powered educational platform with LLM-driven personalization, stateful sessions, and multilingual support